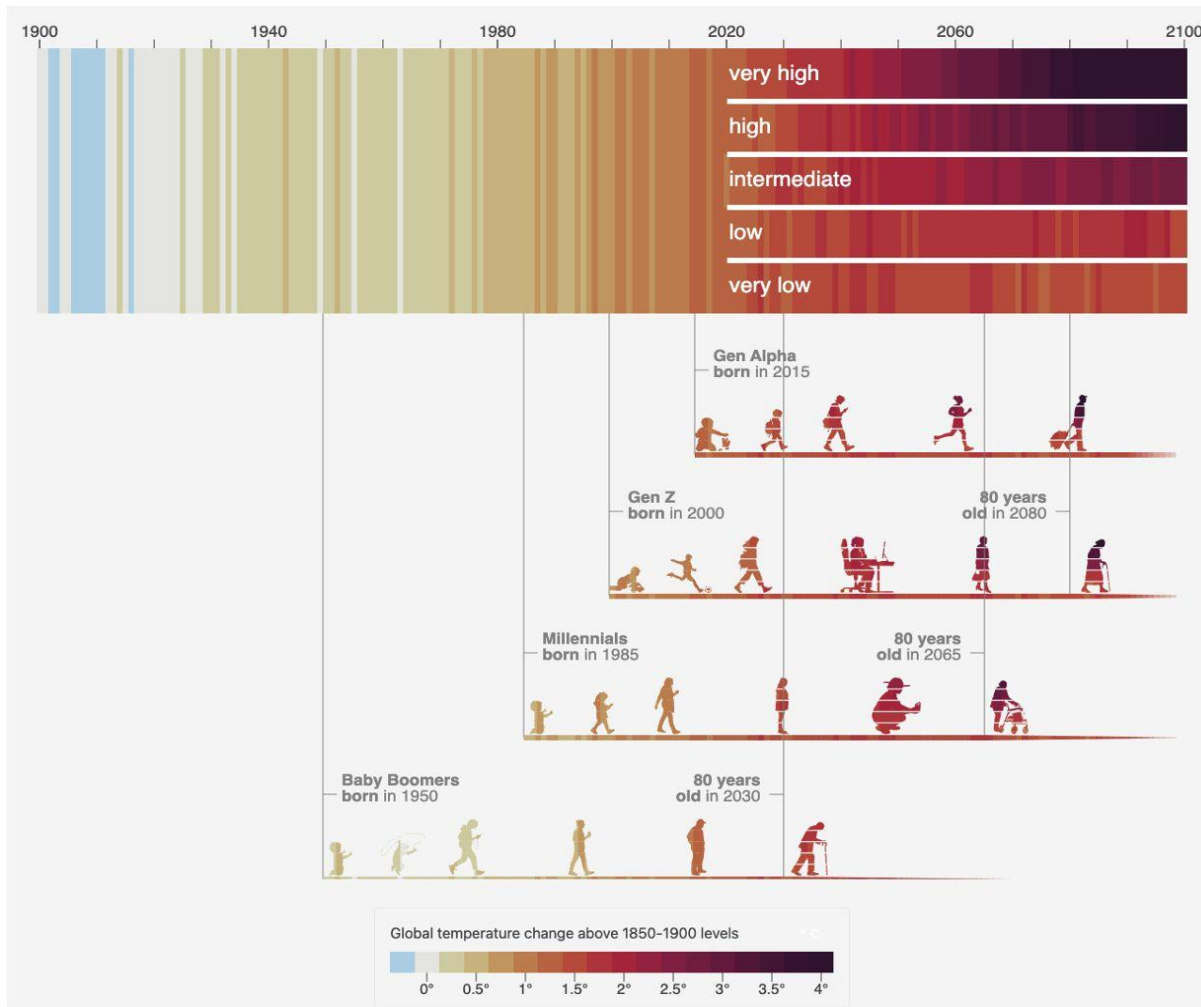


Climate Change Governance: Science, Data & Models



Lecture 9: Introduction to IAMs

ENGG562/ENV562 – Fall 2026

Muhammad Awais

Integrated assessment modeling

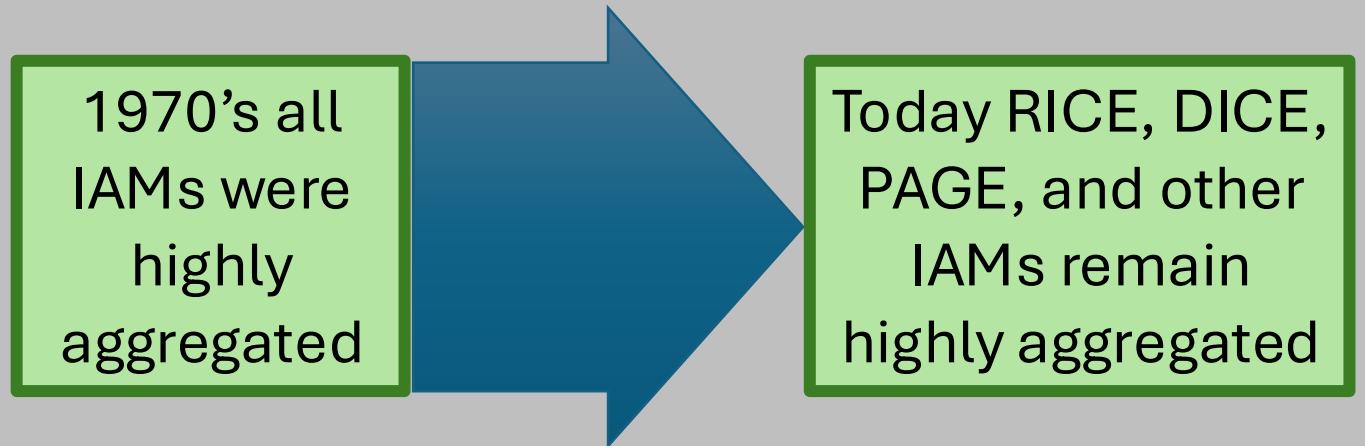
What is integrated assessment modeling?

“integrated assessment is an attempt to combine information, analysis and insights from the physical and social sciences to address the nature of climate change and to develop possible policy responses to it”

John P. Weyant
Climatic Change 95,
p. 317–323, 2009

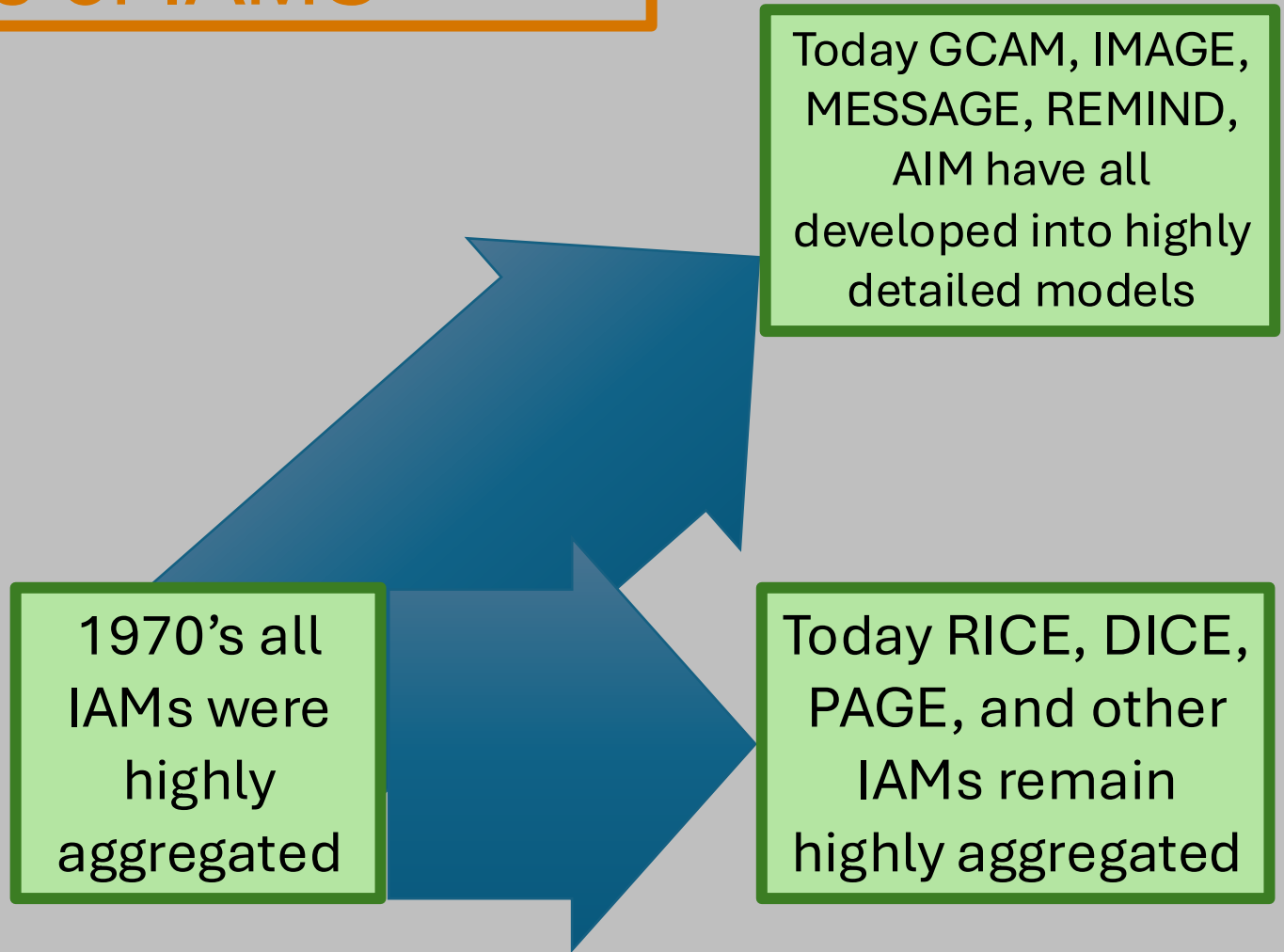
Two Major Types of IAMs

- Integrated assessment models divide into two major branches:
- **Highly Aggregated Models**, where the central problem is determining the **social cost of carbon**, and

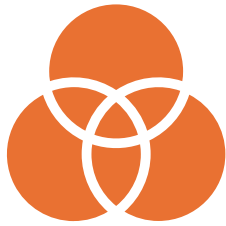


Two Major Types of IAMs

- Integrated assessment models divide into two major branches:
- **Highly Aggregated Models**, where the central problem is determining the **social cost of carbon**, and
- **Highly Detailed Human-Earth System models**, whose primary problem is describing the detailed **interactions of human and physical Earth systems**, in physical units



Two Kinds of Integrated Assessment Models

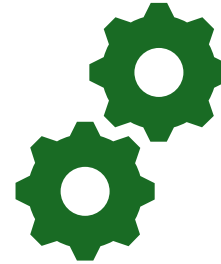


Policy Optimization Models

(aka cost-benefit models)

Focused on finding optimal level of emissions

Usually include impacts at the aggregate level



Policy Evaluation Models

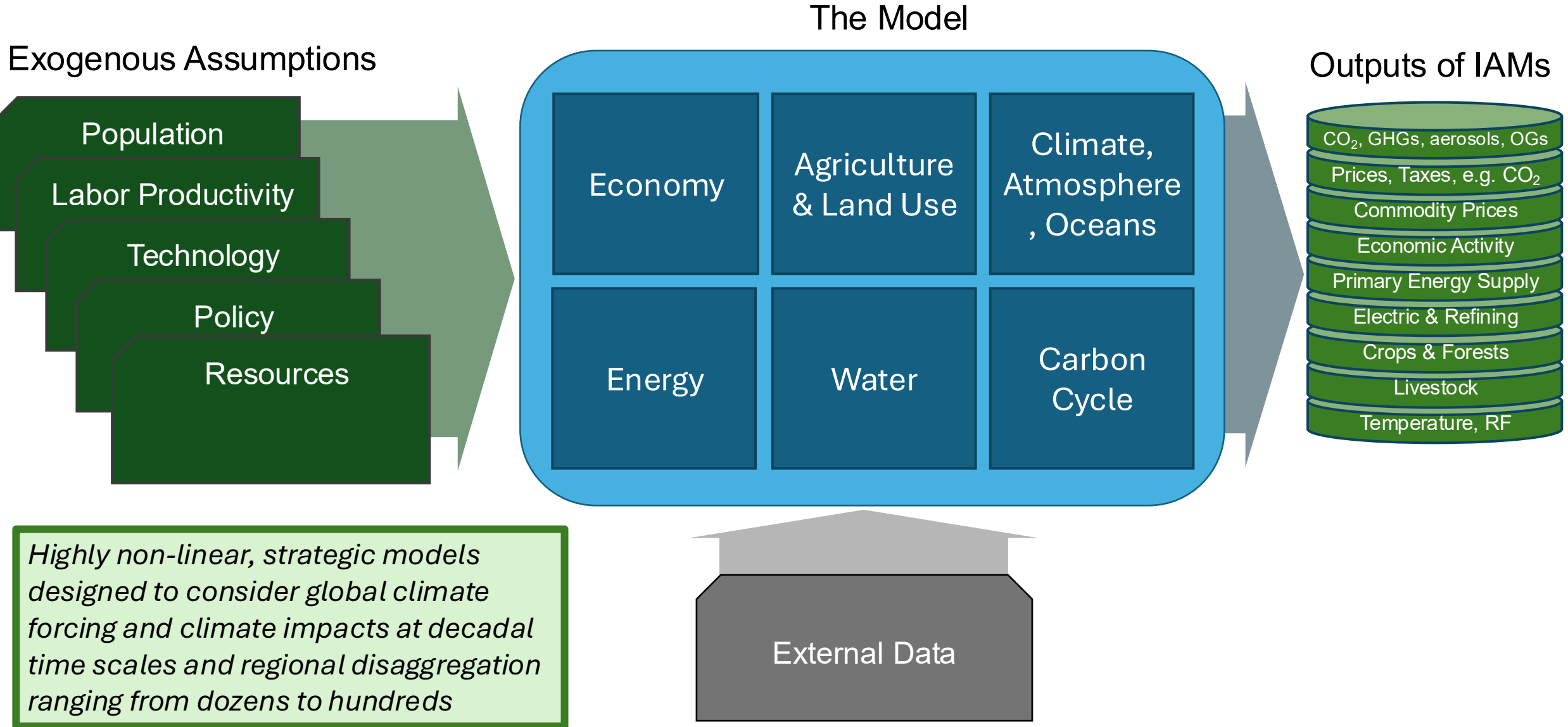
(aka process-based models)

Focused on simulating effects of policies

Usually much more detailed representation of processes and possibly impacts

Can be run backwards – tolerable windows approach

Integrated Assessment Models (IAMs) have external inputs and data, a representation of relevant systems and outputs



Where did IAMs come from?



- Bill Nordhaus started it all
- 1979 *The Efficient Use of Energy Resources*, Yale University Press.
- In 2018, Bill won the Nobel Prize in Economics “for integrating climate change into long-run macroeconomic analysis.”
- That work was followed by the DICE model, which premiered in
- “Future Carbon Dioxide Emissions from Fossil Fuels,” with Gary Yohe, in National Research Council-National Academy of Sciences, ***Changing Climate***, National Academy Press, 1983.
- Dice is highly aggregated. It has 30 equations.
- <http://www.econ.yale.edu/~nordhaus/homepage/Web-DICE-2013-April.htm>

The DICE: the world in 30 Equations

Appendix A: DICE MODEL EQUATIONS

Total factor of Productivity (TFP)

$$A_g(t) = A_{g0} \times e^{[-A_{gd} \times 10 \times (t-1)]} \quad A1$$

$$A(t) = \frac{A(t-1)}{1 - A_g(t-1)} \quad A2$$

Labor

$$L(t) = L_0 \times (1 - L_g(t)) + L_A \times L_g(t) \quad A3$$

$$L_g(t) = \frac{e^{[L_{gv} \times (t-1)]} - 1}{e^{[L_{gv} \times (t-1)]}} \quad A4$$

Capital

$$K(t) = I(t-1) + (1 - \delta_K)^{10} \times K(t-1) \quad A5$$

$$I(t) = s(t) \times Q(t) \quad A6$$

Production Function

$$Y(t) = Y_0 \times \left(\frac{A(t)}{A_0}\right) \times \left(\frac{K(t)}{K_0}\right)^\beta \times \left(\frac{L(t)}{L_0}\right)^{1-\beta} \quad A7$$

$$Q(t) = (\Omega(t) - A(t)) \times Y(t) \quad A8$$

$$Q(t) = C(t) + I(t) \quad A9$$

Climate Change Damage

$$\Omega(t) = \frac{1}{1 + \Psi_1 \times T_{at}(t) + \Psi_2 \times T_{at}^{10}(t)} \quad A10$$

$$T_{at}(t) = T_{at}(t-1) + \xi_1 \times \{F(t) - \xi_2 \times T_{at}(t-1) - \xi_3 \times [T_{at}(t-1) - T_{lo}(t-1)]\} \quad A11$$

$$T_{lo}(t) = T_{lo}(t-1) + \xi_4 \times [T_{at}(t-1) - T_{lo}(t-1)] \quad A12$$

$$F(t) = \Delta R_f \times \log_2 \left(\frac{M_{at}(t)}{M_{at}(1750)} \right) + F_{ex}(t) \quad A13$$

$$M_{at}(t) = E(t-1) + \phi_{11} \times M_{at}(t-1) + \phi_{21} \times M_{up}(t-1) \quad A14$$

$$M_{up}(t) = \phi_{12} \times M_{at}(t-1) + \phi_{22} \times M_{up}(t-1) + \phi_{32} \times M_{lo}(t-1) \quad A15$$

$$M_{lo}(t) = \phi_{23} \times M_{up}(t-1) + \phi_{33} \times M_{lo}(t-1) \quad A16$$

$$\xi_2 = \frac{\Delta R_f}{\Delta T} \quad A17$$

$$E(t) = E_{ind}(t) + E_{land}(t) \quad A18$$

$$E_{ind}(t) = \sigma(t) \times [1 - a(t)] \times Y(t) \quad A19$$

$$E_{land}(t) = E_0 \times 0.9^{t-1} \quad A20$$

$$\sigma(t) = \frac{\sigma(t-1)}{1 - \sigma_g(t)} \quad A21$$

$$\sigma_g(t) = \sigma_{g0} \times e^{[-10 \times \sigma_{gd} \times (t-1) - 10 \times \sigma_{ga} \times (t-1)^2]} \quad A22$$

$$F_{ex}(t) = F_{2000} + (F_{2100} - F_{2000}) \times (t-1) \quad A23$$

$$A(t) = \pi^{(1-\theta_2)}(t) \times \theta_1(t) \times a^{\theta_2}(t) \quad A24$$

Climate Change Abatement

$$\pi(t) = \varphi^{1-\theta_2}(t) \quad A25$$

$$\theta_1(t) = \left(\frac{P_0 \times \sigma(t)}{\theta_2} \right) \times \left(\frac{P_r - 1 + e^{[-P_d \times (t-1)]}}{P_r} \right) \quad A26$$

Utility

$$U(t) = u(t) \times L(t) \quad A27$$

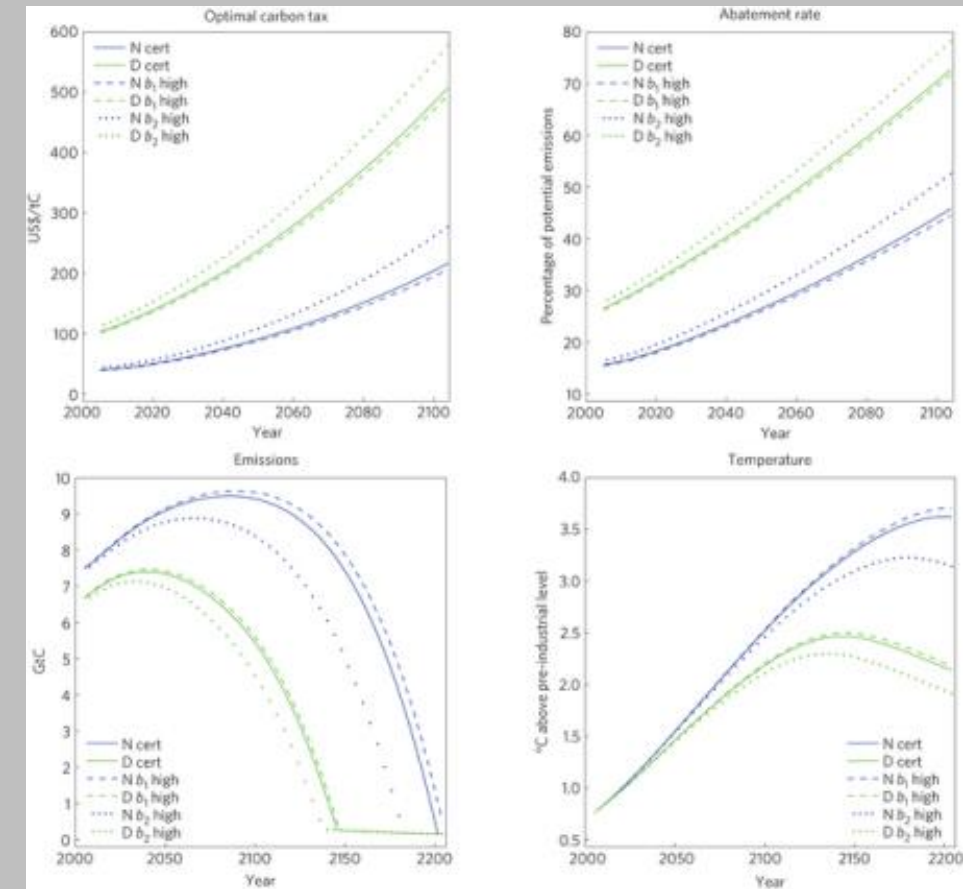
$$u(t) = \frac{c^{1-\alpha}(t)}{1-\alpha} \quad A28$$

$$c(t) = \frac{C(t)}{L(t)} \quad A29$$

$$R(t) = \frac{R(t-1)}{(1+\rho)^{10}} \quad A30$$

Nordhaus central problem is cost-benefit

- What is the socially optimal emissions path?
- What is the optimal global carbon tax?
- What is the optimal change in climate?
- How do you monetize climate damages, many of which do not pass through markets?
- How do you discount costs and benefits?



Source: Crost, Benjamin, and Christian P. Traeger. "Optimal CO2 mitigation under damage risk valuation." *Nature Climate Change* (2014).

The most important contribution highly aggregated IAMs make is defining the Social Cost of Carbon

The social cost of carbon (SCC) is an estimate, in dollars, of the economic damages that would result from emitting one additional ton of greenhouse gases into the atmosphere.

EPA estimates: \$120/tCO₂ (2.5% discount rate),
\$190/tCO₂ (2.0% discount rate),
\$340/tCO₂ (1.5% discount rate).

RFF estimate: \$185/tCO₂ (2.0% discount rate).

Highly aggregated IAMs have drawn the ire of critiques such as Robert Pindyck

These models [IAMs] have crucial flaws that make them close to useless as tools for policy analysis: certain inputs (e.g., the discount rate) are arbitrary, but have huge effects on the SCC estimates the models produce; the models' descriptions of the impact of climate change are completely ad hoc, with no theoretical or empirical foundation; and the models can tell us nothing about the most important driver of the SCC, the possibility of a catastrophic climate outcome. IAM-based analyses of climate policy create a perception of knowledge and precision, but that perception is illusory and misleading.



Researchers such as John Weyant have defended the field of integrated assessment

Arbitrary inputs	Uncertainty is made explicit, not hidden. Multi-model intercomparisons (EMF, IPCC AR6 scenario database) and sensitivity analysis show which assumptions matter and why.
Ad hoc damages	Detailed-process IAMs mostly do cost-effectiveness , not cost-benefit: they ask how to meet a target (e.g., 1.5°C or 2°C) at least cost, which avoids the damage function entirely. Damage estimates are also improving through empirical work (e.g., Burke et al., 2015; Rennert et al., 2022).
Catastrophe ignored	Newer stochastic IAMs incorporate tipping points and tail risk (e.g., Cai, Lontzek and Judd; Dietz et al., 2021), and high tail risk strengthens the case for early action.



Weyant, J. “Some Contributions of Integrated Assessment Models of Global Climate Change” Review Of Environmental Economics And Policy, 2017; 11 (1): 115-137

Researchers such as John Weyant have defended the field of integrated assessment

Researchers such as Weyant have pressed the case for IAMs, and particularly highly detailed IAMs as valuable research and policy support tools.

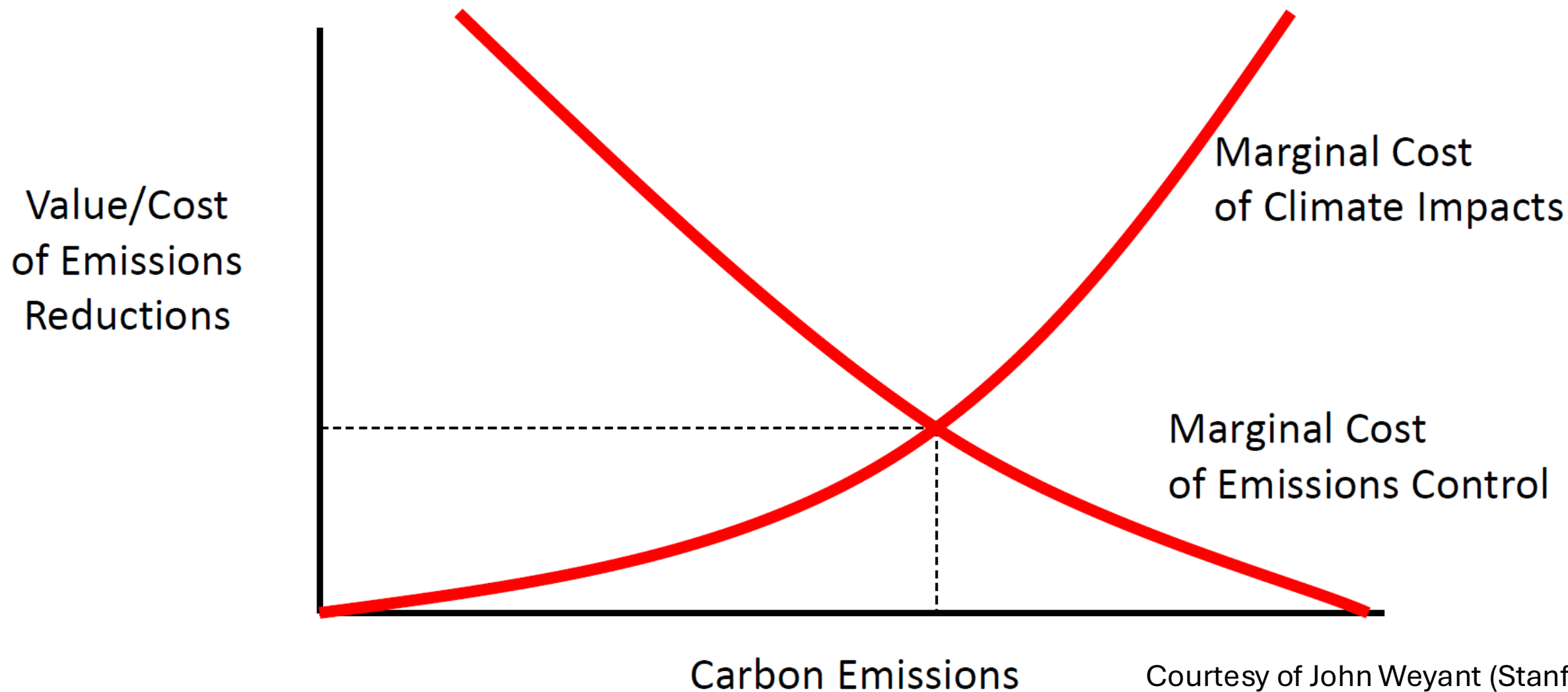
Weyant, J. “Some Contributions of Integrated Assessment Models of Global Climate Change” *Review Of Environmental Economics And Policy*, 2017; 11 (1): 115-137



Cost-Benefit IAMs

How much it costs to reduce emissions vs how much future damage is avoided

Approach: Balancing the Costs of Controlling Carbon Emissions



Courtesy of John Weyant (Stanford)

Generic Types of process-based IAMs

- Process engineering (aka activity analysis) models
 - Represent **individual technologies** (power plants, cars, industries).
 - **Optimize** energy supply to meet demand at *least cost or emissions*.
 - Excellent for technology pathways, mitigation options, and system design.
 - Assume fixed demand, thus limited feedback from markets or economy.
 - Example models: MESSAGEix, TIMES, MARKAL, OSeMOSYS.

Generic Types of process-based IAMs - II

- Energy Market (Partial Equilibrium) Models

These models bring in market behavior — supply, demand, and prices — but focus only on the energy sector.

- Simulate **energy markets**: producers, consumers, and price formation.
- Include feedbacks (e.g., higher prices → lower demand).
- Useful for analyzing carbon taxes, fuel subsidies, energy security.
- Represent energy in moderate detail; treat other sectors simply.
- Example models: POLES, IEA-WEM, NEMS.

Generic Types of process-based IAMs - II

- General Equilibrium Models

These models simulate the entire economy — showing how all sectors interact through markets, production, and consumption.

- Capture **economy-wide feedbacks**: wages, prices, GDP, trade.
- Include households, firms, and government.
- Can evaluate climate policies, carbon pricing, or growth impacts.
- Two main types:
 - CGE (e.g., *EPPA*, *GTAP*, *GCAM*): detailed sector interactions.
 - Optimal Growth (e.g., *REMIND*, *DICE*): long-term welfare optimization.
- Weaker on technology detail and energy engineering realism.

Questions	Type of Process based IAM
A researcher wants to evaluate how different solar and wind technologies can replace coal power at the least cost in Pakistan’s power sector by 2050.	
An economist wants to understand how a global oil price shock would influence GDP, employment, and trade across all economic sectors.	
The government introduces a fuel subsidy reform and wants to analyze how energy demand and fuel prices adjust in response.	
Policymakers want to assess how a ban on imported natural gas would alter domestic energy prices and renewable investment decisions.	

Highly aggregated IAMs have drawn the critiques such as Robert Pindyck

These models [IAMs] have crucial flaws that make them close to useless as tools for policy analysis: certain inputs (e.g., the discount rate) are arbitrary, but have huge effects on the SCC estimates the models produce; the models' descriptions of the impact of climate change are completely ad hoc, with no theoretical or empirical foundation; and the models can tell us nothing about the most important driver of the SCC, the possibility of a catastrophic climate outcome. IAM-based analyses of climate policy create a perception of knowledge and precision, but that perception is illusory and misleading.



Pindyck, Robert S.. 2013. "Climate Change Policy: What Do the Models Tell Us?." *Journal of Economic Literature*, 51(3):860-72.

Researchers such as John Weyant have defended the field of integrated assessment

Researchers such as Weyant have pressed the case for IAMs, and particularly highly detailed IAMs as valuable research and policy support tools.

Weyant, J. “Some Contributions of Integrated Assessment Models of Global Climate Change” *Review Of Environmental Economics And Policy*, 2017; 11 (1): 115-137



Strengths and Limitations

Strengths

- Globally and sectorally comprehensive analysis
- Interlinkages between sectors, regions (incl. trade in some commodities), human and natural systems

Limitations

- Technology focus – integration of behavioral aspects emergent
- Supply side bias – demand side typically less well represented
- Representation of national/sectoral barriers
- Limited spatial and temporal resolution
- Sectoral detail, e.g., IAMs are not power system models

*“The purpose of computing is insights,
not numbers.”*

Richard W. Hamming
Numerical Methods for
Scientists and Engineers
McGraw-Hill, 1962

“Essentially, all models are wrong, but some are useful.”

George E. P. Box
Empirical Model Building
and Response Surfaces
John Wiley & Sons, 1987

An introduction to open, collaborative scientific research

About you...

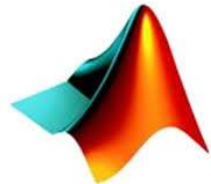
What is your background and experience level with (scientific) programming?



Excel



python™



MATLAB®



GitLab



GitHub

L^AT_EX



G A M S



AMPL

Microsoft Excel as a programming language?

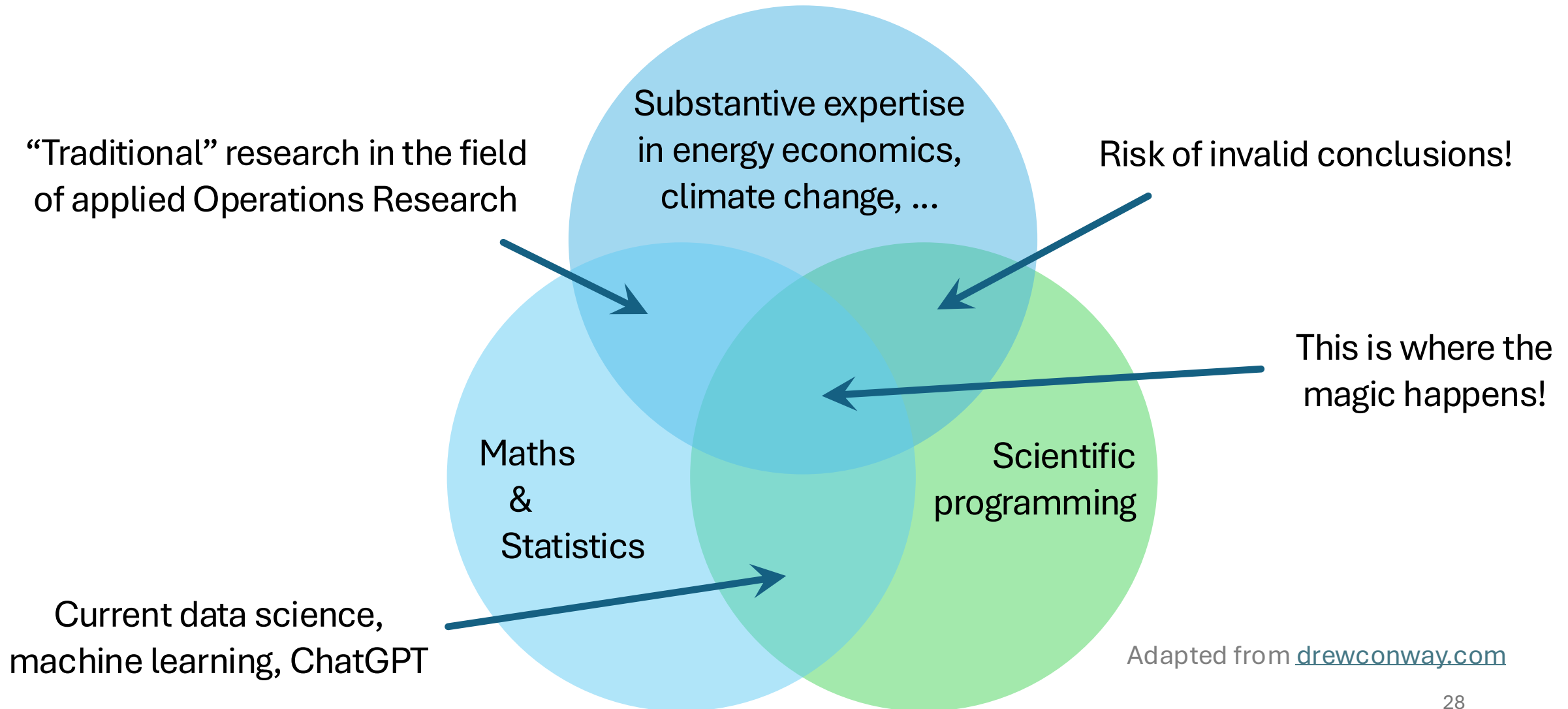
People tend to have strong feelings about Excel...



John Oliver, *Last Week Tonight*, June 5, 2016. Meme from memegenerator.net, Clip on youtube.com

The intersection between energy economics and mathematics

Current research requires substantial expertise in scientific programming



Key misconceptions about best practice in open scientific programming

If you think that this topic is of no concern to you, you're probably wrong

- Who is your main (and usually worst) collaborator?
 - Yourself from six months ago!
And you probably didn't write enough documentation and don't respond to emails
- Why is it a bad idea to use data or software that does not have an open license?
 - Bad karma!
 - Are you intending to distribute your work?
How are you planning to deal with non-open contributions that your project depends on?
- Why should you share data and code under an open-source license?
 - Good karma!
 - Standard licenses have a disclaimer of liability, so you cannot be accountable for problems
 - There is probably a growing expectation from your (potential) collaborators
 - *Treat your GitHub, etc. profile as your "business card" similar to your list of publications*

The FAIR Guiding Principles

Existing digital ecosystem of scholarly data publication prevents us from extracting maximum benefit from our research investments

- Good data management and stewardship is not a goal in itself
 - Rather, it's a pre-condition supporting knowledge discovery and innovation
- Increasingly, science funders, publishers and governmental agencies require data management and stewardship plans for publicly funded research projects
- Digital research objects should be available for **transparency, reproducibility and reusability**
 - This includes data as well as algorithms, tools and workflows to compile and assess data
- Data management must be geared towards human readers *and* machine processing
 - Humans have an intuitive sense of 'semantics' (the meaning or intent of a digital object)
 - But humans are not able to operate at the scope, scale, and speed required for the scale of contemporary scientific data and complexity

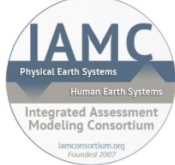
Mark Wilkinson et al. *Scientific Data* 3:160018 (2016) doi: [10.1038/sdata.2016.18](https://doi.org/10.1038/sdata.2016.18)

The distinction between FAIR for humans vs. machines

Humans are good at making sense of visual elements...



International Institute for
Applied Systems Analysis
IIASA www.iiasa.ac.at



IAMC
Physical Earth Systems
Human Earth Systems
Integrated Assessment
Modeling Consortium
iamconsortium.org
Founded 2002

IAMC 1.5°C Scenario Explorer and Data hosted by IIASA

Daniel Huppmann, Elmar Kriegler, Volker Krey, Keywan Riahi, Joeri Rogelj, Steven K. Rose, John Weyant, Nico Bauer, Christoph Bertram, Valentina Bosetti, Katherine Calvin, Jonathan Doelman, Laurent Drouet, Johannes Emmerling, Stefan Frank, Shinichiro Fujimori, David Gernaat, Arnulf Grubler, Celine Guivarch, Martin Haigh, Christian Holz, Gokul Iyer, Etsushi Kato, Kimon Keramidas, Alban Kitous, Florian Leblanc, Jing-Yu Liu, Konstantin Löffler, Gunnar Luderer, Adriana Marcucci, David McCollum, Silvana Mima, Alexander Popp, Ronald D. Sands, Fuminori Sano, Jessica Strefler, Junichi Tsutsui, Detlef Van Vuuren, Zoi Vrontisi, Marshall Wise, Runsen Zhang. **IAMC 1.5°C Scenario Explorer and Data hosted by IIASA.** *International Institute for Applied Systems Analysis & Integrated Assessment Modeling Consortium.* (2018) [10.22022/SR15/08-2018.15429](https://doi.org/10.22022/SR15/08-2018.15429)

Item Type: Dataset



Archive
iamc15_scenario_data_all_regions_r1.1.xlsx - Published Version (Release 1.1)
iamc15_scenario_data_all_regions_r1.xlsx - Published Version (Release 1.0)

Please access this resource from the [Scenario Explorer Website](#). For usage rights please see our license [here](#).

Version History:

Release 1.1 (February 7, 2019)
This release includes additional timeseries data to increase reproducibility of the figures and tables in the SR15, and it corrects a number of data issues identified since Release 1.0. None of the changes have any impact on the assessment in the SR15.

Release 1.0 (October 15, 2018)
Scenario ensemble release for the soft launch of the IPCC SR15 following the approval plenary in Incheon, Republic of Korea.

Please view the [About page](#) for details.

Rendered version of the landing page for doi [10.22022/SR15/08-2018.15429](https://doi.org/10.22022/SR15/08-2018.15429)

The distinction between FAIR for humans vs. machines

Computers require additional structure to parse information

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<h1>IAMC 1.5°C Scenario Explorer and Data hosted by IIASA</h1>
<Creator:Author>Daniel Huppmann</Creator:Author>
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hosted by IIASA</b></CreationName:Title>
<Agent:Publisher><i>International Institute for Applied Systems Analysis
& Integrated Assessment Modeling Consortium</i></Agent:Publisher>.
<DateOfPublishing>(2018)</DateOfPublishing>
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<a target="_blank" href="https://doi.org/10.22022/SR15/08-2018.15429">
10.22022/SR15/08-2018.15429</a></Name:Identifier:DoiName>
Item Type: <Type>Dataset</Type>
Please access this resource from the <Digital:Website><a target="_blank"
href="https://data.ene.iiasa.ac.at/ iamc-1.5c-explorer">
<b>Scenario Explorer Website</b></a></Digital:Website>.
```

Source code of landing page for doi [10.22022/SR15/08-2018.15429](https://doi.org/10.22022/SR15/08-2018.15429)

The FAIR Guiding Principles (II)

Scientific work should be Findable, Accessible, Interoperable and Reusable

Data and/or metadata...

Findable	• F1. ... are assigned a unique and persistent identifier (Digital Object Identifier, DOI) • F2. ... are described with rich metadata (defined by R1 below) • F3. ... clearly and explicitly include the identifier of the data it describes
Accessible	• F4. ... are registered or indexed in a searchable resource (including Google) • A1. ... are retrievable by their identifier using a standardized protocol
Interoperable	• A2. ... are accessible, even when the data are no longer available • I1. ... use a formal, shared, applicable language for knowledge representation • I2. ... use vocabularies that follow FAIR principles
Reusable	• I3. ... include qualified references to other (meta)data • R1. ... are richly described with a plurality of accurate and relevant attributes: clear data license, detailed provenance, meet community standards

A one-page guide to open & FAIR practices

Five best-practice steps to make your research open & FAIR_{v1.0}



You may think that putting your work* on a website already makes it free & open. But that's not quite true – follow these steps to implement best practice of **#openscience!**

* data sets, text, tables, figures & illustrations, source code, scientific software, ... even #Horizon2020 deliverables

1. Open

If you want your *work to be read, used & shared by others*, be explicit about it: For text, data, figures, ... – use the [CC-BY license](#) | For code, visit [choosealicense.com](#)

2. Findable

To make it easy for others to find and cite your work, get a [digital object identifier \(DOI\)](#) and add a *recommended citation*

3. Accessible

Depositing your work in an institutional repository or a service like [zenodo](#) ensures that your work is still *available even after the end of the project*

4. Interoperable

Using established community standards, data formats and software packages lets others *quickly understand and use your work*

5. Reusable

To make it easy for others to *build on your work*, make sure to assign a version number and relevant (machine-readable) metadata



This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 835896

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Please cite as: Daniel Huppmann et al., 2020
Five best-practice steps to make your research open & FAIR v1.0
doi: [10.22022/ene/04-2020.16404](https://doi.org/10.22022/ene/04-2020.16404) | url: openENTRANCE.eu

The rationale for proper version control tools

In love and in scientific research, there is no such thing as “final”...



Adapted from “notFinal.doc” at “Piled Higher and Deeper” by Jorge Cham, <http://phdcomics.com>

Best Practices

- When writing code
 - Write software for people, not computers
 - Don't repeat yourself
 - Make it correct, then make it fast
 - Make incremental changes
 - Use consistent style

- To be reproducible
 - Automate repetitive tasks
 - Use version control
 - Plan for mistakes
- Working as a team
 - Document design, purpose, and assumptions
 - Conduct code reviews
 - Use available release & management tools

Required reading and preparation

- Preparation for scientific programming exercises in this course:
 - create a [GitHub](#) account
 - know what it means to 'clone' a repository, make a 'commit' and 'push'
 - either get familiar with 'git' using the command line
 - or get familiar with a program for working with git repos (for novice users, try [Gitkraken](#))
 - install Python (for novice users, I recommend that you use [Anaconda](#))
 - get familiar with the basic Python syntax
- Required reading:
 - The FAIR Guiding Principles, Mark Wilkinson et al. *Scientific Data* 3:160018 (2016)
doi: [10.1038/sdata.2016.18](https://doi.org/10.1038/sdata.2016.18)
 - Greg Wilson et al. Good enough practices in scientific computing. *PLOS Computational Biology* 13(6), 2017. doi: [10.1371/journal.pcbi.1005510](https://doi.org/10.1371/journal.pcbi.1005510)

GitHub repository

- Create a GitHub user account:
<https://github.com/>

The screenshot shows the GitHub interface for a repository named "ENG426-526-CC_Data_Science_Models". The repository is private and has 0 stars, 0 forks, and 0 watchers. It is currently on the "main" branch, which has 2 branches and 0 tags. The repository was last committed 2 months ago by user "awais307". The commit history shows 3 commits, with the most recent one being a merge pull request #1 from LUMS-WIT/dev. The repository contains several files: .gitignore, Climate_Change_Governance_Syll..., LICENSE, resources.ipynb, and rubrics.ipynb. The README section is currently empty, with a placeholder text "Add a README". The repository is licensed under the MIT license.

ENG426-526-CC_Data_Science_Models Private

main 2 Branches 0 Tags

Go to file t + <> Code

awais307 Merge pull request #1 from LUMS-WIT/dev 3472a4f · 2 months ago 3 Commits

File	Commit	Time
.gitignore	add rubrics and resuources	2 months ago
Climate_Change_Governance_Syll...	add rubrics and resuources	2 months ago
LICENSE	Initial commit	2 months ago
resources.ipynb	add rubrics and resuources	2 months ago
rubrics.ipynb	add rubrics and resuources	2 months ago

README MIT license

Add a README

About

This is the course repository for course on ENG426/526

- MIT license
- Activity
- Custom properties
- 0 stars
- 0 watching
- 0 forks

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No releases published
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